

for agrarian economies such as India. Proof-of-concepts have been set up around Europe already.

Machine learning has thus impacted a far broader set of sectors than just computer science.

Future technologies

While there is considerable interest surrounding ML and AI, I will attempt to walk you through this territory, leaving out the hype. From a human perspective, it is quite plausible to accidentally create artificial general intelligence that could impact all of humanity. However, we are still years away from such a possible misfortune.

Machine learning has some exciting new developments and future plans. The launch of the ONNX format that allows the interchange of deep neural networks across different libraries, serves as a prelude to the cooperation that is needed to ensure quick, easy and meaningful experiences for all. It is envisioned to take on a central role as research and development of models progresses.

Reinforcement learning is a largely unexplored space that is currently seeing much activity. Examples include deep-q networks, playing Atari and other video games, or learning to walk, run, and jump in a virtual environment.

Truly self-driving vehicles (with Level 5 autonomy), that are still limited to the design desk, should enter the market after being subjected to an extensive set of safety regulations, given the past incidents with Uber, Tesla, Waymo and others.

Quantum AI is a relevant subject area that could take off in the near future, though in its current state it is restricted mainly to theories and hypotheses. It could offer tremendous speed-ups and reinvent the complete pipeline that currently relies on training models on CPUs or GPUs.

Improvements in unsupervised learning may come about as strategies for clustering improve in efficiency. Similar progress will come about in defining the evaluation metrics for clustering that usually relies on

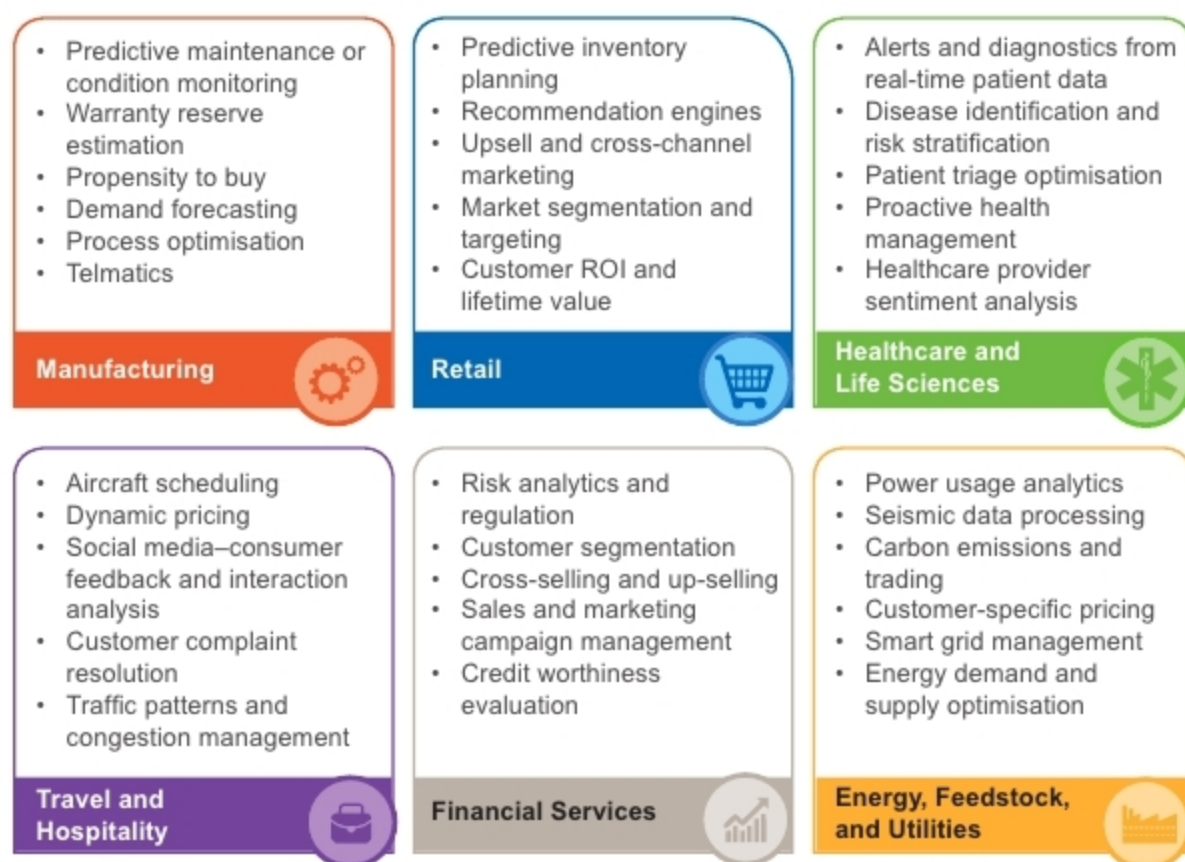


Figure 3: The applications of machine learning (source: *forbes.com*)



Figure 4: Machine learning is here to stay!

assessment of its peers as a group.

Ethics and AI policy come to the fore when you give a company access to all your data, since there is a huge potential for this to be misused. End-to-end encryption, security, anonymity and abstraction of data then gain importance. With the European General Data Protection Regulation rules kicking in, it has forced businesses to rethink their business practices and explicitly seek permission before using our data.

Personalisation will run deeper than usual, with data profiling at the root of

the user's 'profile' based on likes, views and browsing patterns.

Reproducibility and interpretability are major concerns, especially when treating neural networks as black boxes and seeking to explain the results of experiments, or when attempting to replicate large scale training without the original datasets. Attempting to derive intuitive information about the intermediary steps, learned features and other aspects of the training of a neural network is not easy.

The brain-computer interface will soon be realised, though may be not perfectly. Emulating the constructs underpinning an actual human brain was the original motivation behind developing neural networks; so only the future can tell when this will materialise.

In the years to come, it is possible that we will be able to create some semblance of general intelligence. This can change the world or even destroy it—let's just keep our fingers crossed that we will all survive! **END** 🐧

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